

# PHYSICAL LAYOUT PORTFOLIO

Analog and Mixed-Signal  
VLSI Design

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"FOCUSING ON SILICON AREA  
OPTIMIZATION, DRC/LVS  
COMPLIANCE, AND  
PARASITIC EXTRACTION FOR  
CUSTOM INTEGRATED  
CIRCUITS."

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# Technical Portfolio: IC Layout & Verification

This compendium showcases custom physical designs implemented on the **Cadence Virtuoso** platform. It highlights proficiency in translating schematics into DRC (Design Rule Check) and LVS (Layout vs Schematic) clean mask data, with a specific focus on matching techniques and area efficiency.

## CORE COMPETENCIES FOCUSED IN THIS REPORT

|                      |                                  |                            |
|----------------------|----------------------------------|----------------------------|
| Custom Mask Design   | DRC / LVS Verification           | Common-Centroid Matching   |
| Parasitic Extraction | Guard Ring & Latch-up Prevention | Standard Cell Optimization |

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| Precision analog macro design featuring differential pair interdigitation and common-centroid matching to minimize offset and process gradients.               |                                                     |           |

\*Note: All layout designs are verified using industrial standard verification decks (DRC/LVS).

# Layout Project 1 — Schmitt Trigger Layout

CMOS Schmitt Trigger Layout (Hysteresis Comparator Cell in 45nm CMOS)

## Overview

Designed and implemented the transistor-level layout of a CMOS Schmitt Trigger in Cadence Virtuoso using GPDK45 technology, followed by extracted transient verification to analyze hysteresis behavior under parasitic loading.

## Design Objectives

- Realize fabrication-compliant custom CMOS Schmitt Trigger layout
- Preserve hysteresis behavior after parasitic extraction
- Optimize transistor placement for compact area and routable interconnect
- Achieve clean DRC/LVS verification
- Evaluate extracted transient response for RC-induced delay effects

## Layout Methodology

The Schmitt Trigger layout was implemented using custom transistor placement and compact routing strategies:

- Logical grouping of pull-up/pull-down feedback devices
- Compact interconnect routing to reduce parasitic loading
- Shared diffusion where applicable to reduce area
- Proper well and substrate contact placement for robust operation
- Input/output pins aligned for integration into higher-level blocks

## Layout Snapshots

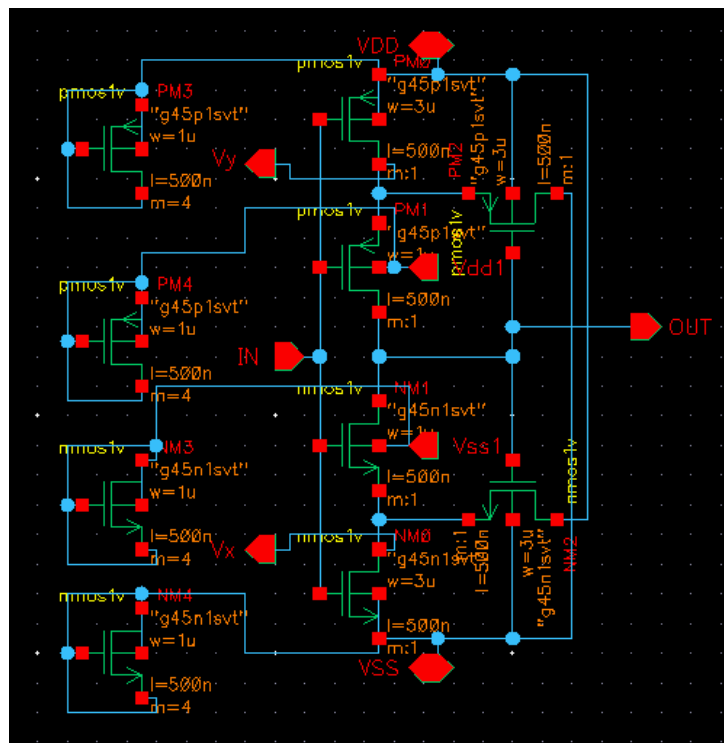


Figure 1- Schematic of Schmitt Trigger

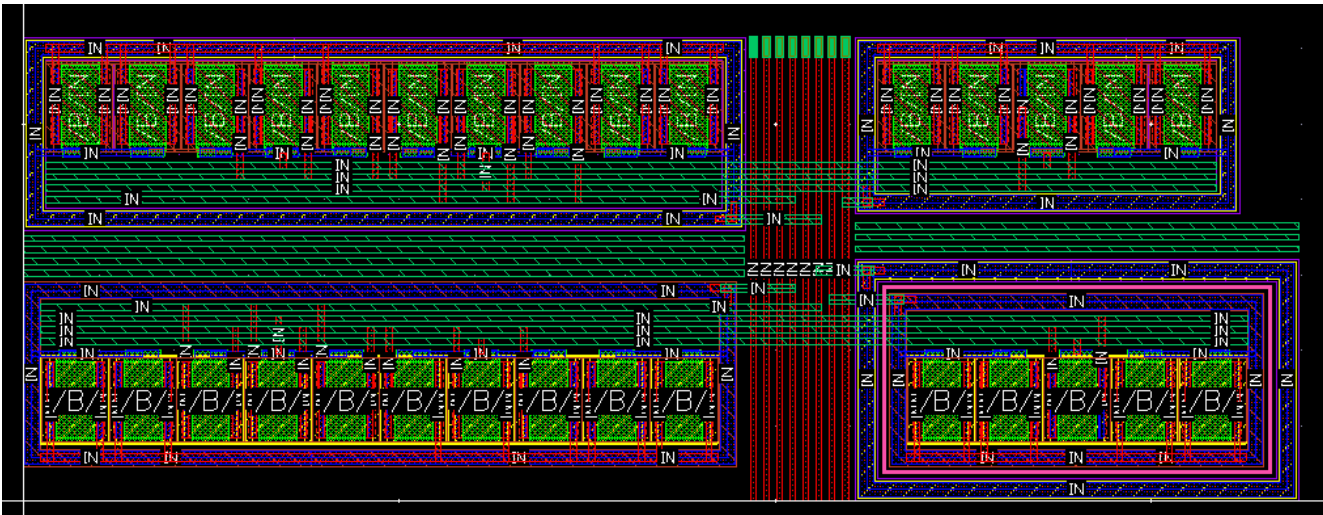


Figure 2- Layout of Schmitt Trigger

## Verification

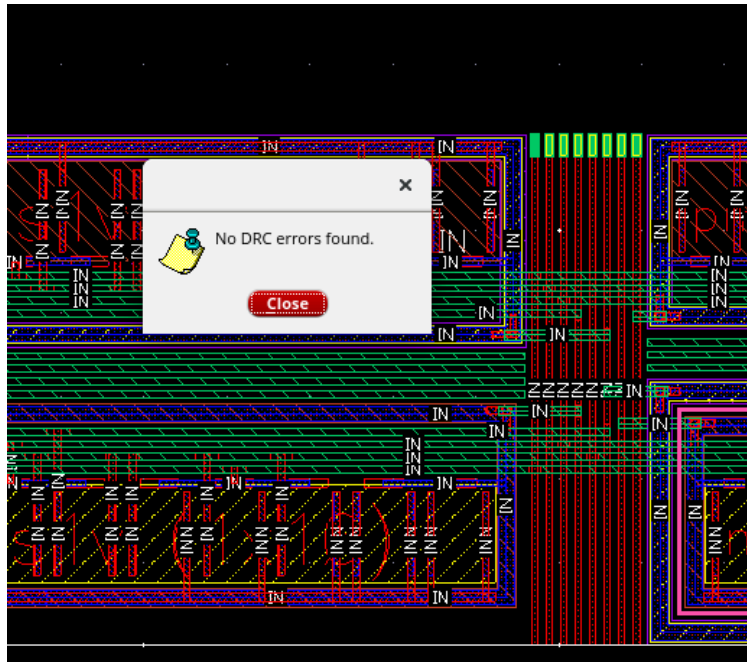


Figure 3- Clean DRC of Schmitt Trigger

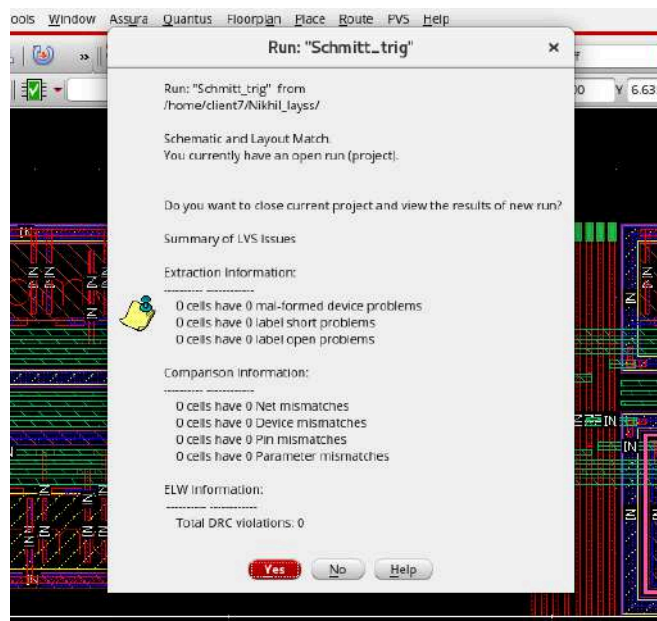


Figure 4- Clean LVS of Schmitt Trigger

- DRC Status: Clean
- LVS Status: Matched

## Post-Layout Extraction

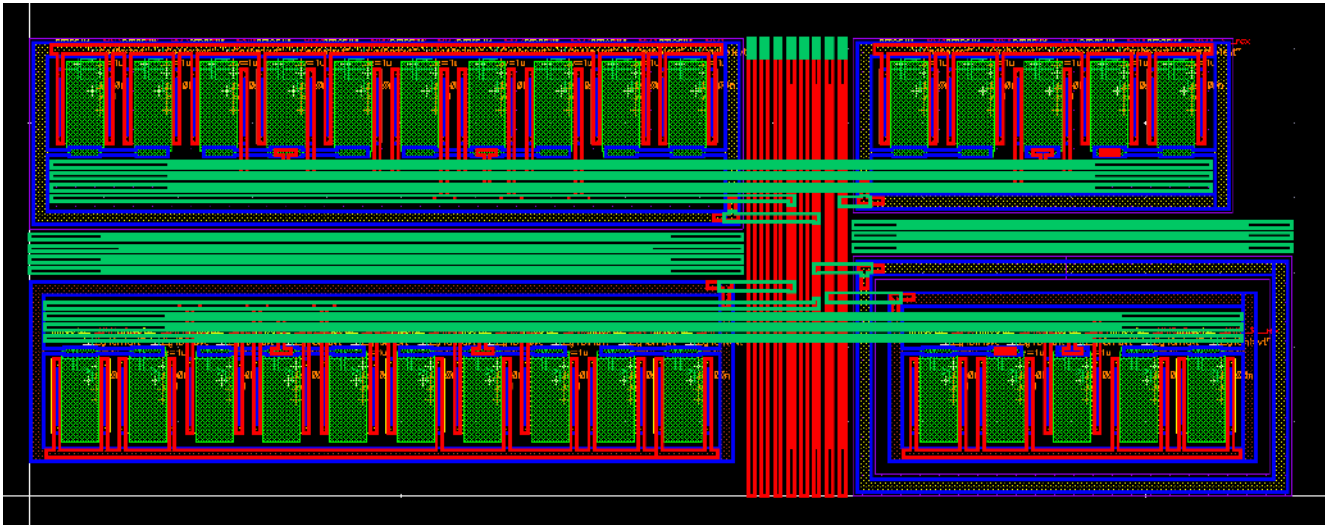


Figure 5- AV Extracted of Schmitt Trigger

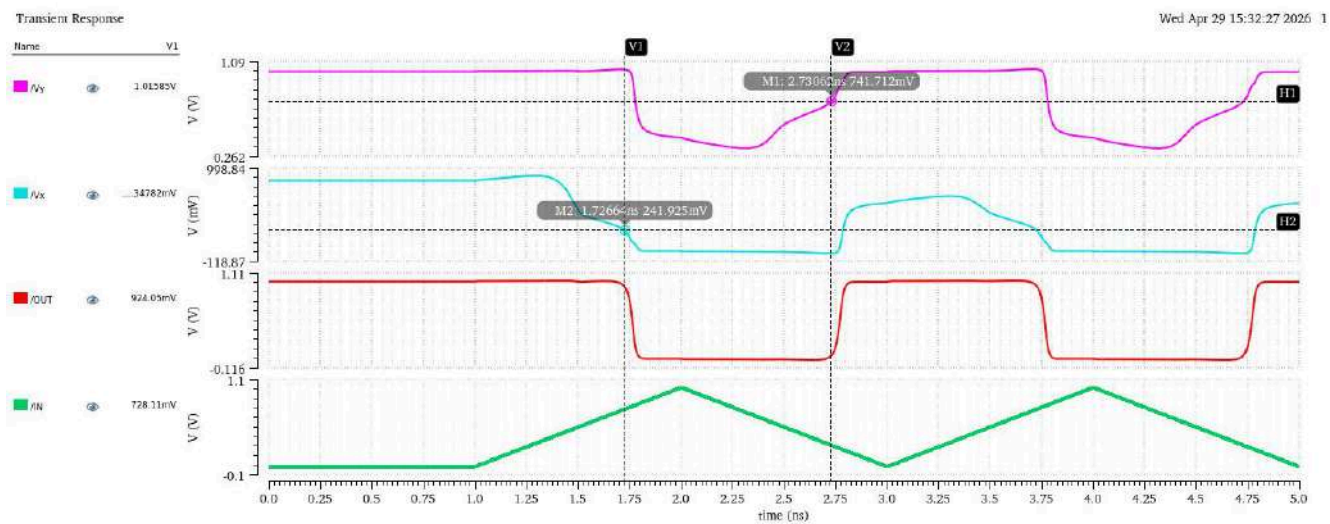


Figure 6- Transient analysis of Schmitt Trigger

## Results

| Parameter         | Value                             |
|-------------------|-----------------------------------|
| Technology        | 45 nm                             |
| DRC Violations    | 0                                 |
| LVS Mismatches    | 0                                 |
| Layout Area       | 104 $\mu\text{m}^2$               |
| Verification Type | AV Extracted Transient Simulation |

## **Design Insights**

- Post-layout extracted simulation confirmed preservation of Schmitt Trigger hysteresis behavior under parasitic loading
- Compact routing minimized excessive RC degradation in feedback paths
- Feedback transistor placement strongly influences switching threshold symmetry and hysteresis window
- Layout parasitics introduce additional propagation delay but do not affect functional correctness

## **Summary**

Successfully implemented and verified a custom CMOS Schmitt Trigger layout in 45nm technology, demonstrating understanding of feedback-based transistor layouts, parasitic-aware verification, and custom cell physical design methodology.

# Layout Project 2 — JK Flip-Flop Layout

## JK Flip-Flop Standard Cell Layout with Post-Layout Parasitic Analysis (180nm CMOS)

### Overview

Designed and implemented the physical layout of a CMOS JK Flip-Flop in Cadence Virtuoso using GPDK180 technology through hierarchical composition of NAND-based standard cells, followed by parasitic extraction and post-layout timing/power analysis.

### Design Objectives

- Realize fabrication-compliant sequential logic layout
- Maintain compact hierarchical standard-cell structure
- Preserve functional behavior after parasitic extraction
- Quantify post-layout timing and power degradation
- Achieve clean DRC/LVS verification

### Layout Methodology

The JK Flip-Flop was implemented using a hierarchical standard-cell based layout methodology:

- NAND gate cells abutted to form complete flip-flop structure
- Uniform cell-height maintained across sub-blocks
- Continuous VDD/GND rails used for standard-cell compatibility
- Feedback routing minimized to reduce parasitic delay
- Boundary-aligned I/O pins placed for hierarchical integration

### Layout Snapshots

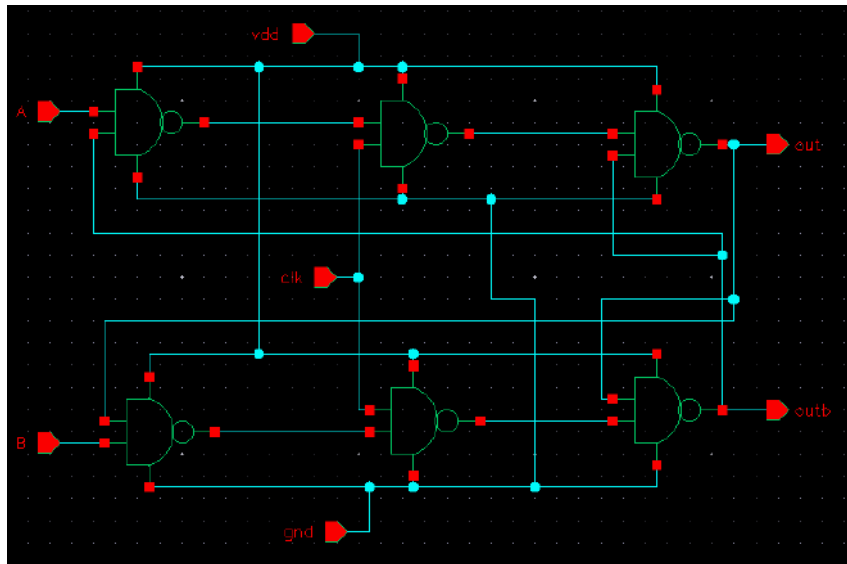


Figure 1- Schematic of JK Flip Flop



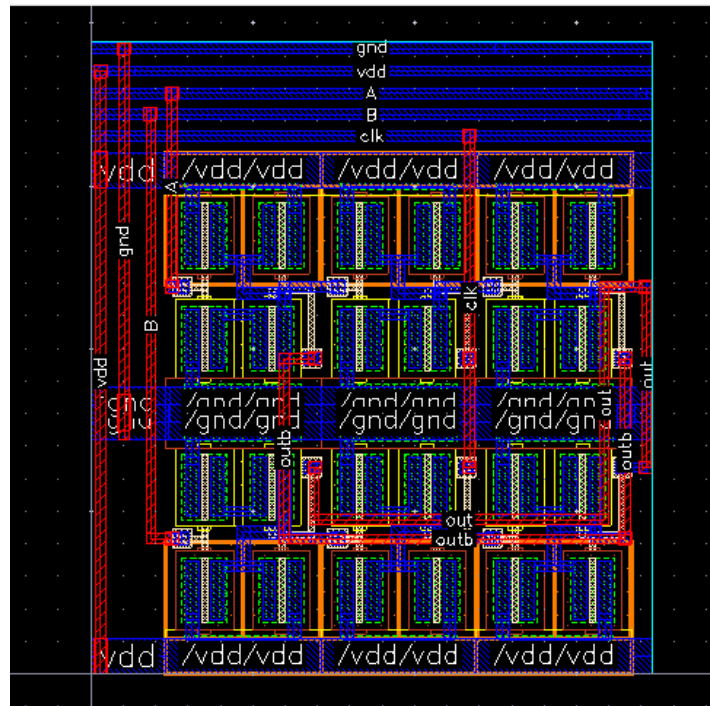


Figure 2- Layout of JK Flip Flop

## Verification

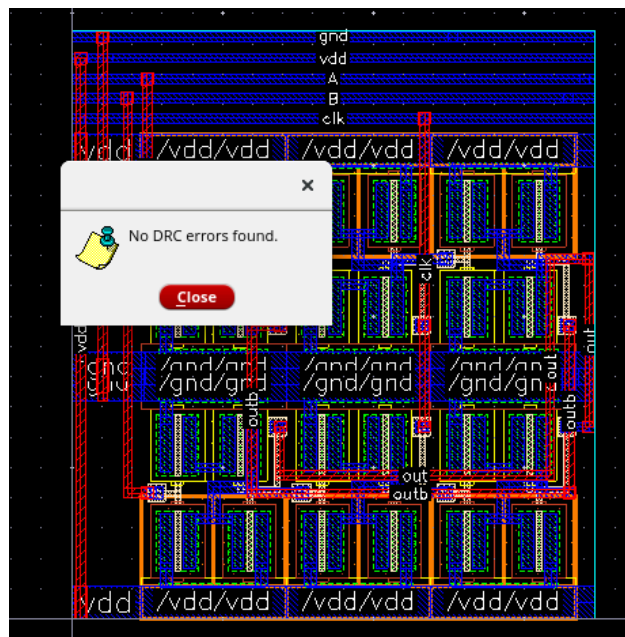
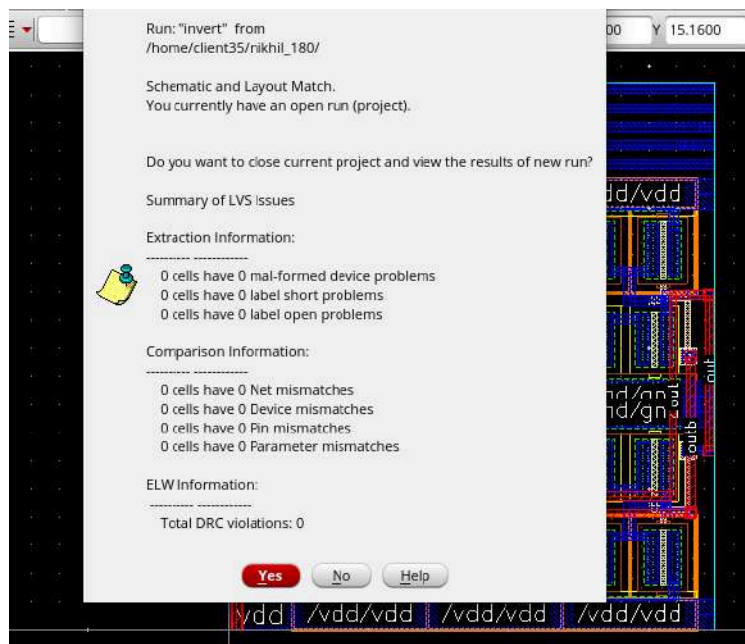


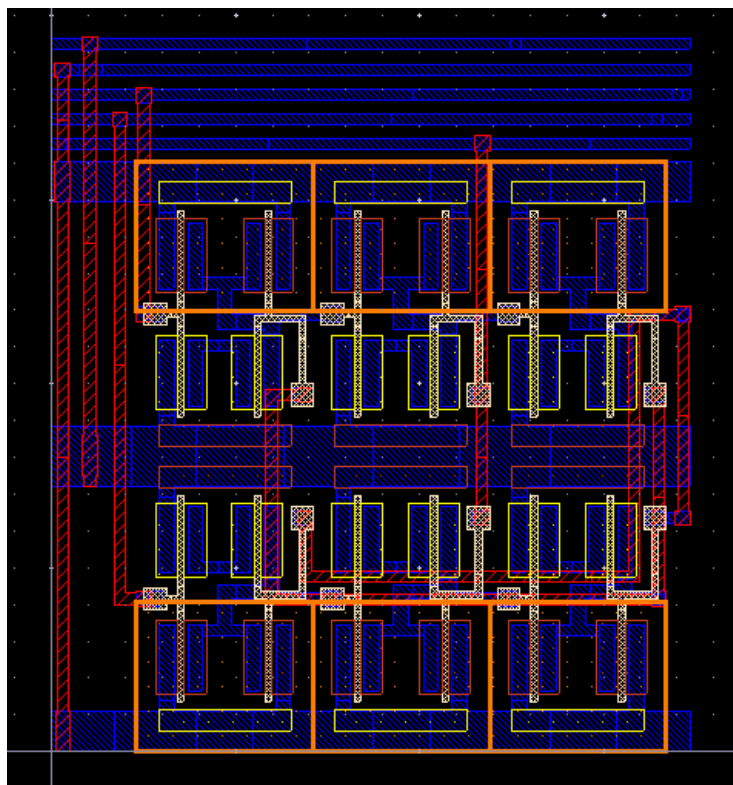
Figure 3- DRC Clean of JK Flip Flop





- DRC Status: Clean
- LVS Status: Matched

## Post-Layout Extraction



*Figure 5- AV Extracted Layout View*

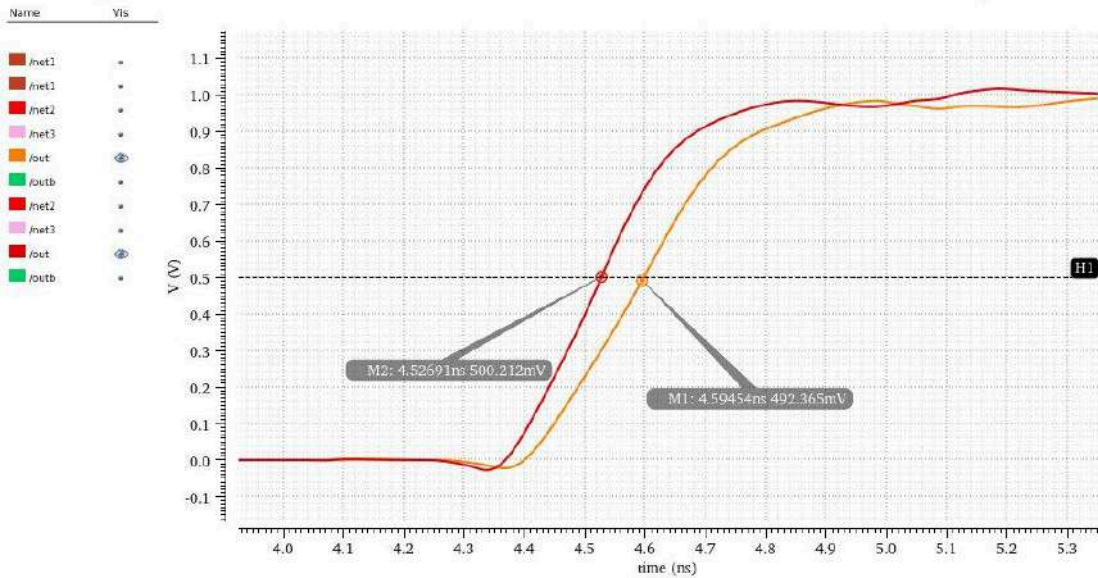


Figure 6- Rising Edge difference b/w schematic v/s av extracted layout

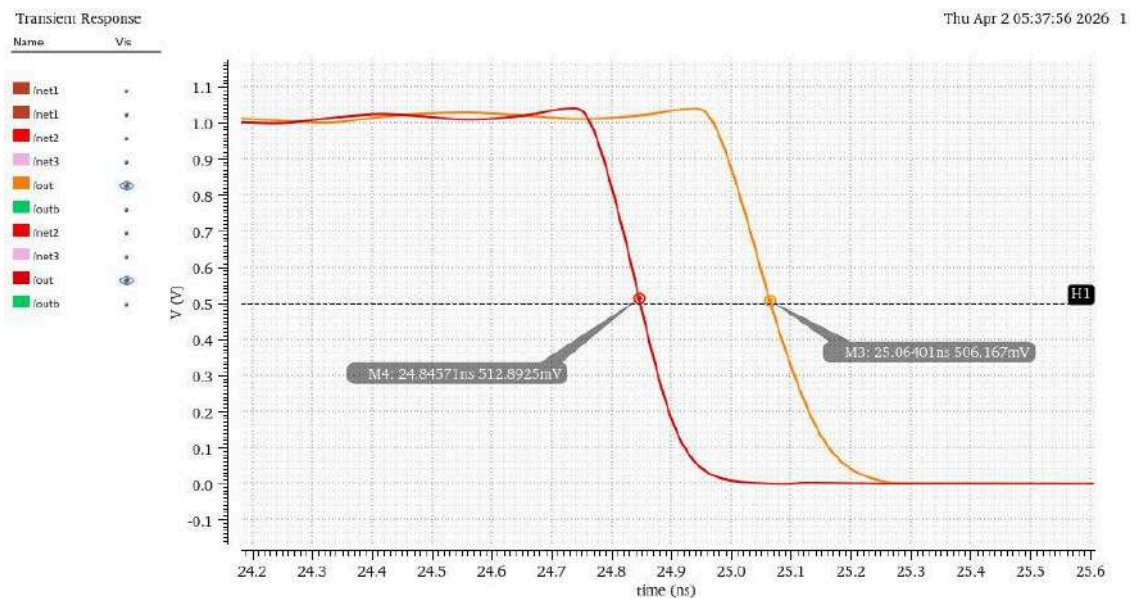


Figure 7- Falling Edge difference b/w schematic v/s av extracted layout

## Results

| Parameter                | Value                 |
|--------------------------|-----------------------|
| Technology               | 180 nm                |
| Layout Area              | 336.7 $\mu\text{m}^2$ |
| DRC Violations           | 0                     |
| LVS Mismatches           | 0                     |
| Rising Edge Delay Shift  | ~60 ps                |
| Falling Edge Delay Shift | ~210 ps               |
| Schematic Power          | 13.49 $\mu\text{W}$   |
| AV Extracted Power       | 16.09 $\mu\text{W}$   |

## **Design Insights**

- Post-layout extraction revealed measurable delay degradation due to RC parasitics in interconnect and diffusion
- Falling edge experienced larger degradation, indicating asymmetric parasitic loading in pull-down network
- Extracted power increased due to additional parasitic capacitance charging/discharging
- Hierarchical standard-cell methodology enabled compact and structured sequential logic implementation

## **Summary**

Successfully designed and verified a hierarchical JK Flip-Flop layout with post-layout parasitic analysis, demonstrating practical understanding of sequential standard-cell physical design and extracted timing/power characterization.

# Layout Project 3 — 6T SRAM Bitcell Layout

6T SRAM Bitcell Layout with Stability and Read/Write Characterization (90nm CMOS)

## Overview

Designed and implemented a compact 6T SRAM bitcell layout in Cadence Virtuoso using GPDK90 technology, followed by functional characterization of stability, read/write delay, and power to evaluate memory-cell robustness.

## Design Objectives

- Achieve compact high-density SRAM bitcell implementation
- Maintain symmetry for matched storage-node behavior
- Optimize layout for array-compatible replication
- Ensure fabrication-compliant DRC/LVS clean verification
- Evaluate memory stability and read/write performance

## Layout Methodology

The SRAM bitcell was implemented using conventional 6T architecture:

- Two cross-coupled CMOS inverters for storage
- Two access transistors for bitline access

Key layout strategies:

- Symmetrical placement of cross-coupled inverter devices
- Matched internal storage-node routing
- Shared diffusion to minimize cell footprint
- Compact wordline/bitline routing for array compatibility
- Boundary pin placement aligned for tiling in memory arrays

## Layout Snapshots

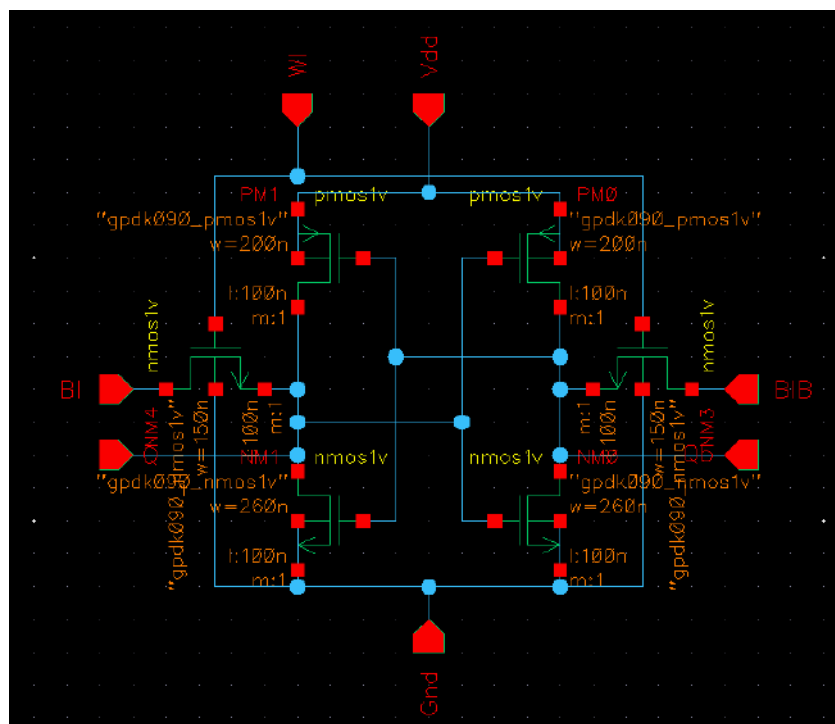


Figure 1- Schematic of 6T SRAM

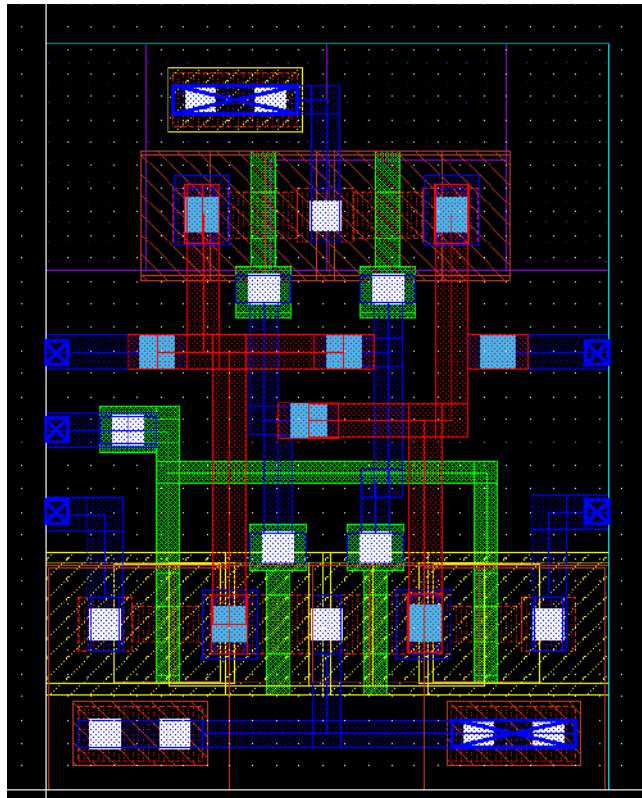


Figure 2- Layout of 6T SRAM

## Verification

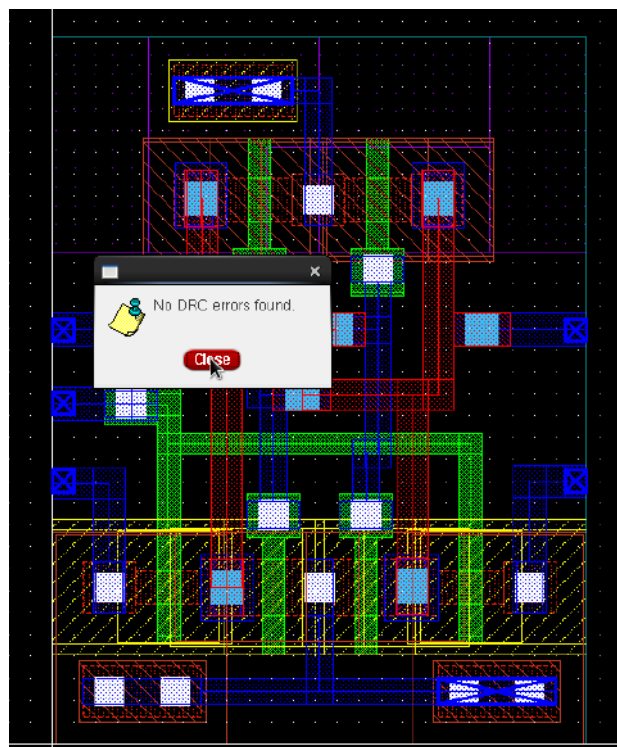


Figure 3- Clean DRC of 6T SRAM

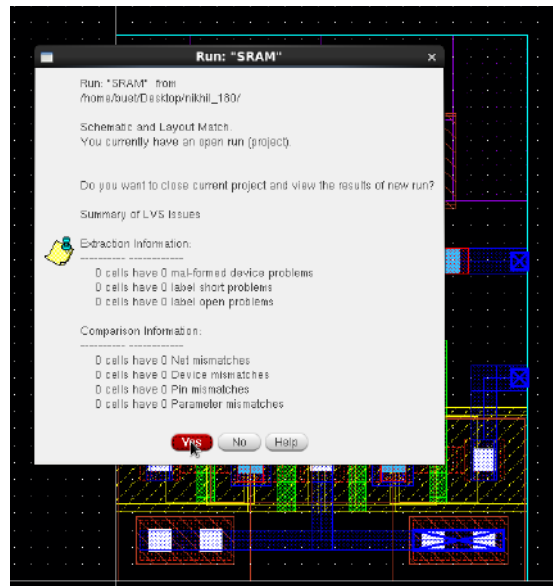


Figure 4- Clean LVS of 6T SRAM

- DRC Status: Clean
- LVS Status: Matched

## Characterization

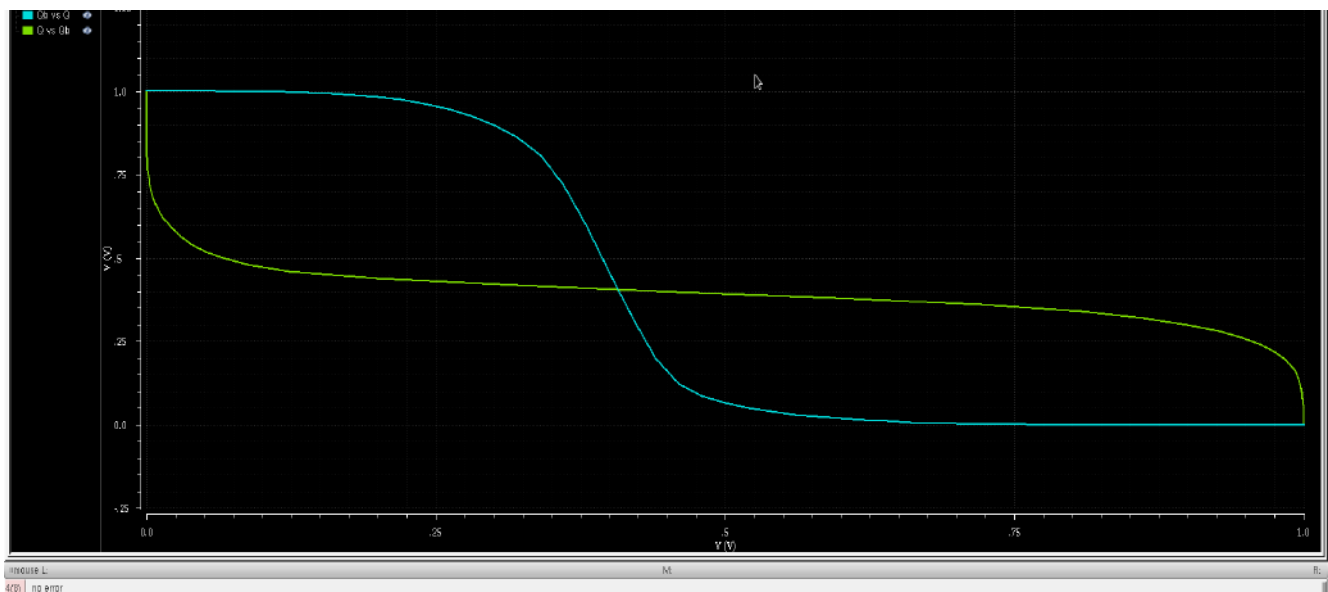


Figure 5- Butterfly curve of 6T SRAM

## Results

| Parameter              | Value                |
|------------------------|----------------------|
| Technology             | 90 nm                |
| Layout Area            | 7.88 $\mu\text{m}^2$ |
| Hold SNM               | ~200 mV              |
| Worst-Case Write Delay | 26.23 ps             |
| Read Delay             | 5.21 ps              |
| DRC Violations         | 0                    |
| LVS Mismatches         | 0                    |

## **Design Insights**

- Symmetric bitcell layout is critical for balanced storage-node behavior and robust read/write operation
- Hold SNM of 180 mV indicates stable data retention during standby operation
- Write delay exceeds read delay due to stronger feedback contention during overwrite operation
- Compact diffusion sharing improves density while preserving array-compatible structure

## **Summary**

Successfully implemented and characterized a compact 6T SRAM bitcell layout, demonstrating understanding of dense memory-cell physical design, symmetry constraints, and SRAM stability/performance tradeoffs.



# Layout Project 4 — OTA Full Custom Analog Layout

Operational Transconductance Amplifier (OTA) Full Custom Analog Layout with Post-Layout AC Verification (180nm CMOS)

## Overview

Designed and implemented the full custom analog layout of a CMOS Operational Transconductance Amplifier (OTA) in Cadence Virtuoso using GPDK180 technology, followed by post-layout extracted AC analysis to evaluate gain and frequency-response degradation due to parasitics.

## Design Objectives

- Realize fabrication-compliant full custom analog layout
- Preserve matching and symmetry across critical analog devices
- Minimize parasitic imbalance in differential signal paths
- Maintain functional performance after extraction
- Achieve clean DRC/LVS verification

## Layout Methodology

The OTA layout was implemented using analog-aware physical design practices:

- Symmetrical placement of differential input pair
- Matched routing of critical differential nets
- Common-centroid / interdigitated placement where applicable
- Dummy devices inserted for edge matching
- Guard rings added for substrate noise isolation
- Compact current mirror and bias network placement to reduce mismatch/parasitics

## Layout Snapshots

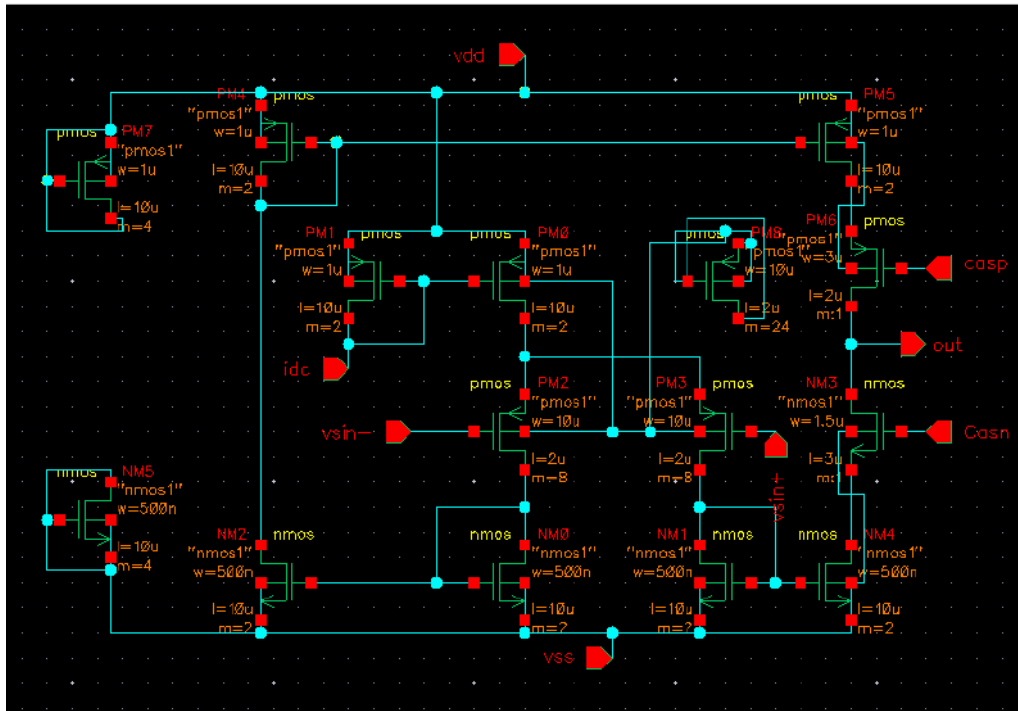


Figure 1- Schematic of OTA

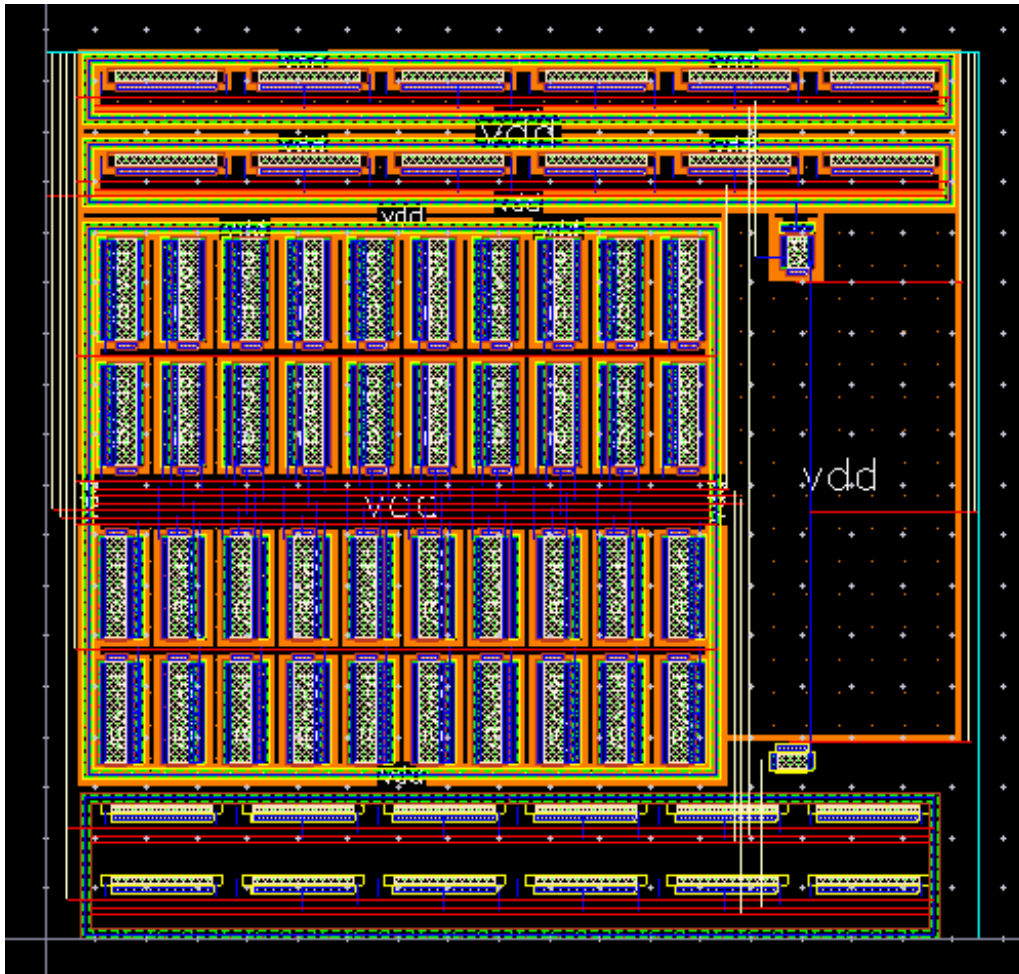


Figure 2- Layout of OTA

## Verification

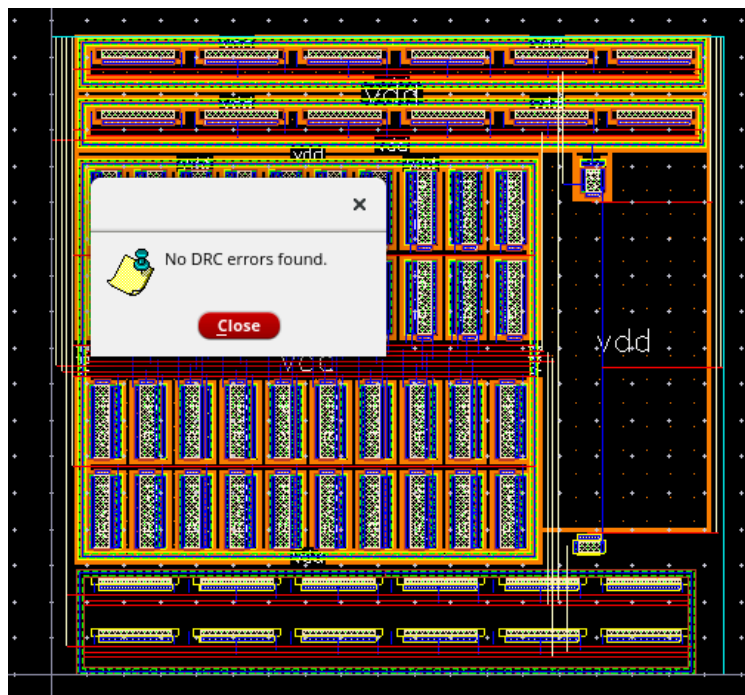


Figure 3- Clean DRC of OTA

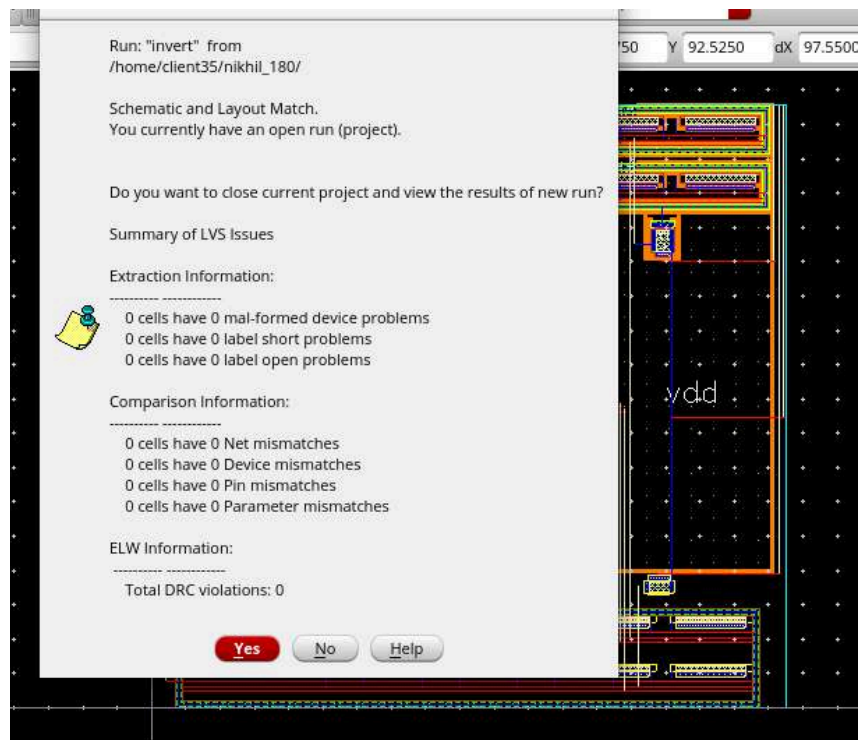


Figure 4- Clean LVS of OTA

- DRC Status: Clean
- LVS Status: Matched

## Post-Layout Extraction

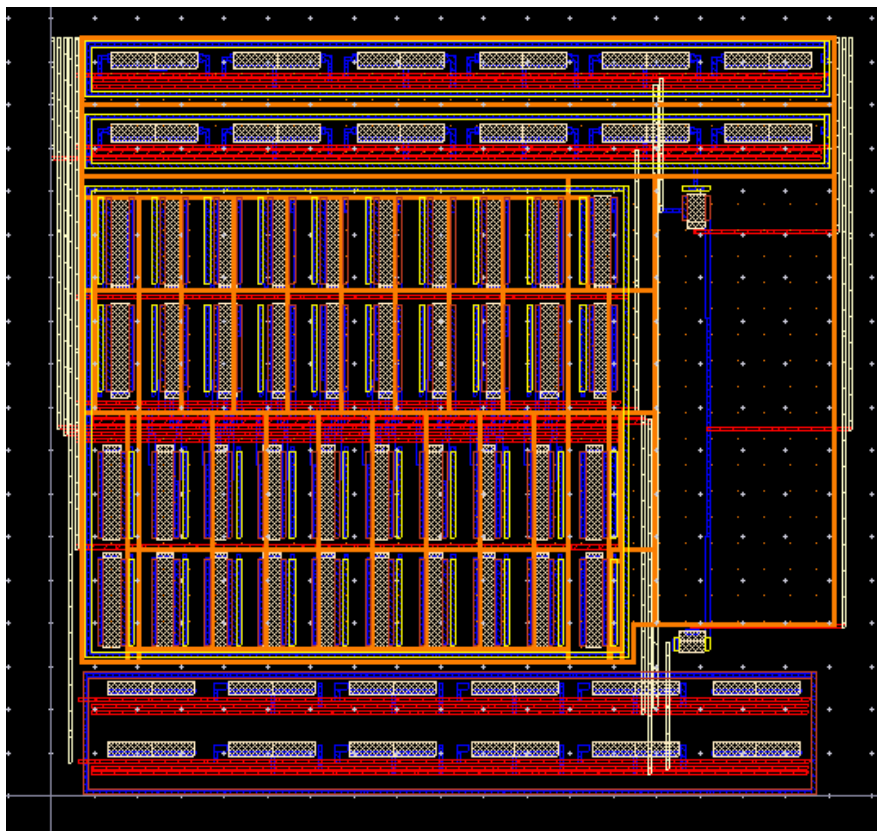


Figure 5- AV Extracted view of OTA

# Transient Response

Mon May 4 05:07:00 2026 1

Name Vis

■ /out  
■ /out

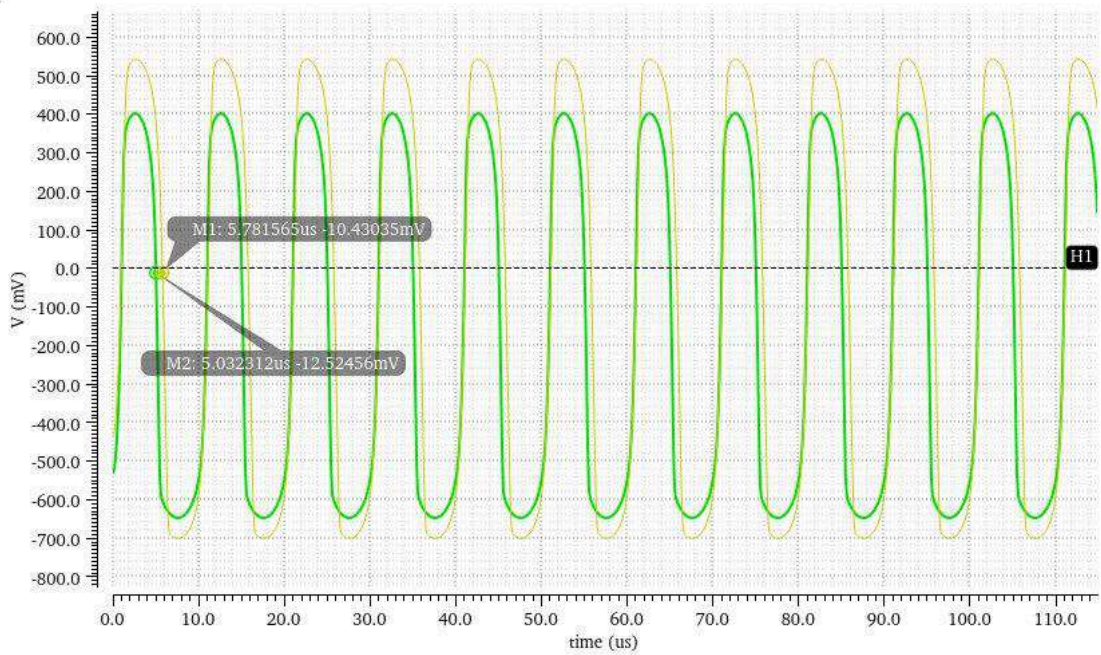


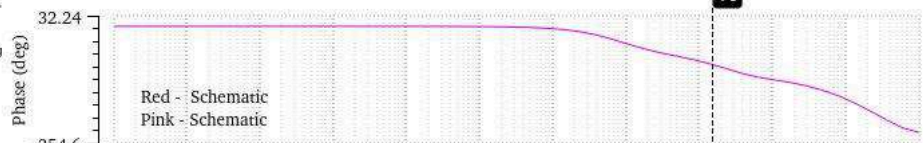
Figure 6- Transient response of Schematic vs Layout of OTA

# AC Response

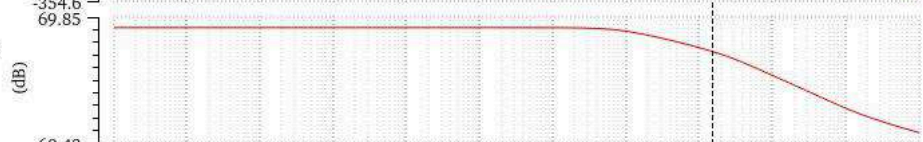
Mon May 4 05:11:12 2026 1

Name V1

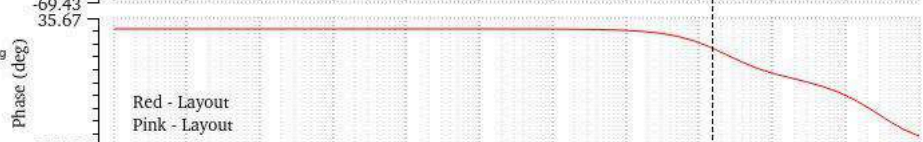
■ ...nwrapped(VF("/out")/VF("/net5"))



■ dB20(VF("/out")/VF("/net5"))



■ ...nwrapped(VF("/out")/VF("/net5"))



■ dB20(VF("/out")/VF("/net5"))

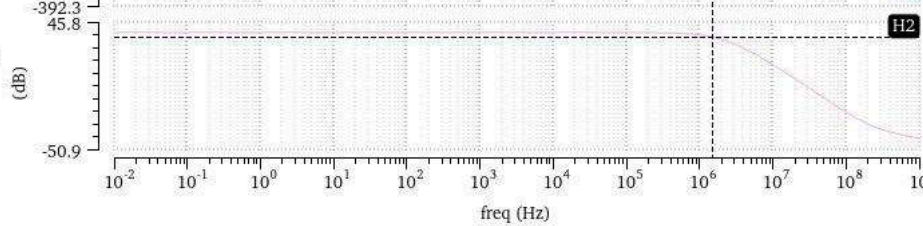


Figure 6- Gain and Phase response of Schematic and Layout of OTA



Results

| Parameter         | Value                         |
|-------------------|-------------------------------|
| Technology        | 180 nm                        |
| Layout Area       | 8.13 nm <sup>2</sup>          |
| DRC Violations    | 0                             |
| LVS Mismatches    | 0                             |
| Verification Type | Post-Layout AC Analysis       |
| Extracted Metrics | Gain / PM / BW / GBW Verified |

The following table quantifies the shift between idealized schematic performance and the physical implementation after **RC Extraction (av\_extracted)**.

| Parameter            | Schematic Value | Post-Layout Value | Deviation |
|----------------------|-----------------|-------------------|-----------|
| DC Gain ( $A_v$ )    | 58.2 dB         | 38.4 dB           | -34.0%    |
| Phase Margin (PM)    | 136.2°          | 62.57°            | -54.0%    |
| 3-dB Bandwidth (BW)  | 120 kHz         | 1.5 MHz           | +1150%    |
| Gain Bandwidth (GBW) | 97 MHz          | ~106 MHz          | +9.2%     |

Technical Observations & Root Cause Analysis

Based on the extracted results and overlaid waveforms, the following factors were identified as primary causes for performance variations:

- **Transient Amplification Discrepancy:** The transient waveform (Yellow) appears more amplified than the schematic (Green). This suggests that while the **low-frequency DC Gain dropped**, the circuit developed a **peaking behavior** at higher frequencies. This is typically caused by parasitic inductance or inductive-like behavior in the feedback/bias network, which creates a "boost" in magnitude before the final roll-off.
- **Gain Degradation (19.8 dB Drop):** The significant reduction in DC gain is attributed to high parasitic resistance ( $R_p$ ) in the signal path. **IR drops** across narrow metal interconnects reduced the effective transconductance ( $g_m$ ) of the input differential pair.
- **Pole Shift and Bandwidth Expansion:** The shift in 3-dB bandwidth (from 120 kHz to 1.5 MHz) indicates a reduction in total output resistance ( $R_{out}$ ) caused by parasitic conductance paths introduced during the layout process.
- **Stability Degradation:** The sharp reduction in Phase Margin (from 136° to 62.5°) confirms that parasitic capacitance ( $C_p$ ) introduced a secondary pole much closer to the unity-gain frequency.

## Key Engineering Insights Gained

1. **Idealized Targets vs. Physical Reality:** Confirmed that schematic-level simulations provide an "idealized" target; only **av\_extracted** simulations capture the true physical constraints of metal resistance and substrate coupling.
2. **Parasitic Sensitivity:** Learned that analog blocks are exponentially more sensitive to layout geometry than digital logic. A "DRC-clean" layout can still fail performance specs if parasitics are not actively managed.
3. **Routing Optimization:** Realized that critical high-gain signal paths require wider metal tracks to minimize  $R_p$  and increased spacing between sensitive nodes to reduce  $C_p$ .
4. **Verification-Driven Design:** Discovered that debugging post-layout discrepancies is a vital skill, allowing for refinements that ensure success before fabrication.

## Design Insights

- Symmetric differential placement minimized systematic mismatch in input pair devices
- Matched routing reduced parasitic imbalance across differential paths
- Guard rings improved substrate isolation for sensitive analog nodes
- Post-layout AC analysis captured RC-induced degradation in gain/bandwidth characteristics
- Analog performance proved highly sensitive to routing parasitics and device proximity

## Summary

This implementation demonstrates proficiency in identifying and characterizing the impact of physical parasitics on circuit functionality. By diagnosing the gain-bandwidth shifts and transient amplification changes, this project underscores a commitment to **Parasitic-Aware Design**.